AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

5.Configure Apache Web Server on Linux Instance in AWS

SRILEKA B

CSE-Artificial Intelligence & Machine Learning,

Sri Eshwar College of Engineering,

Coimbatore.

Objective

EC2 (Elastic Compute Cloud) is a core serviceprovided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache,FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

  Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

Used Services in AWS

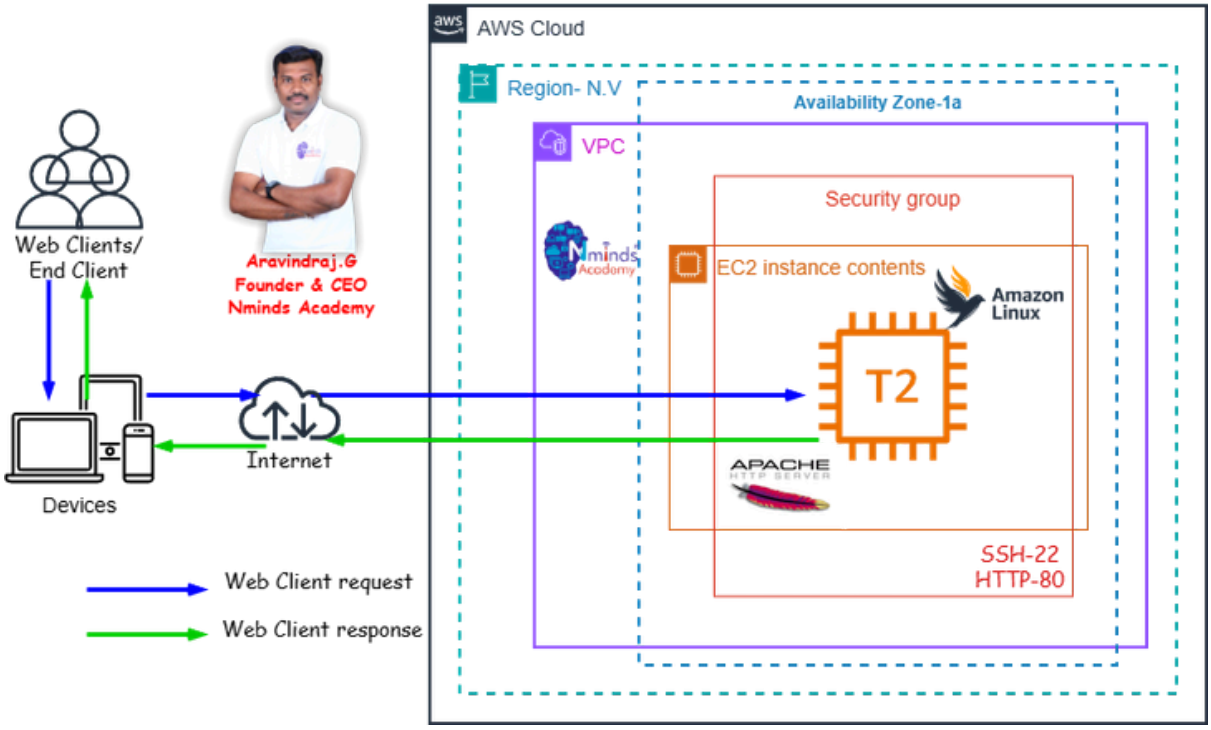
Amazon EC2

Used Services in Linux Server

Apache Webserver

Topology

How to Configure the Apache web server in Linux Server with AWS EC2



Execution Tasks:

```
root@ip-172-31-45-237:~
 Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
 Installing       : mod_lua-2.4.66-1.amzn2023.0.1.x86_64
 Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
 Installing       : httpd-2.4.66-1.amzn2023.0.1.x86_64
 Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64
 Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64
 Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64
 Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
 Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
 Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
 Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
 Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
 Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch
 Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64
 Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64

nstalled:
 apr-1.7.5-1.amzn2023.0.4.x86_64                          apr-util-1.6.3-1.amzn2023.0.2.x86_64
 apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
 generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch        httpd-2.4.66-1.amzn2023.0.1.x86_64
 httpd-core-2.4.66-1.amzn2023.0.1.x86_64                  httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
 httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                 libbrotli-1.0.9-4.amzn2023.0.2.x86_64
 mailcap-2.1.49-3.amzn2023.0.3.noarch                     mod_http2-2.0.27-1.amzn2023.0.3.x86_64
 mod_lua-2.4.66-1.amzn2023.0.1.x86_64

omplete!
root@ip-172-31-45-237 ~]#
```

```
root@ip-172-31-45-237:~
 apr                          x86_64          1.7.5-1.amzn2023.0.4          amazonlinux          129 k
 apr-util                     x86_64          1.6.3-1.amzn2023.0.2          amazonlinux           97 k
 apr-util-lmdb                x86_64          1.6.3-1.amzn2023.0.2          amazonlinux           13 k
 generic-logos-httpd          noarch          18.0.0-12.amzn2023.0.3        amazonlinux           19 k
 httpd-core                   x86_64          2.4.66-1.amzn2023.0.1         amazonlinux          1.4 M
 httpd-filesystem             noarch          2.4.66-1.amzn2023.0.1         amazonlinux           13 k
 httpd-tools                  x86_64          2.4.66-1.amzn2023.0.1         amazonlinux           81 k
 libbrotli                    x86_64          1.0.9-4.amzn2023.0.2          amazonlinux          315 k
 mailcap                      noarch          2.1.49-3.amzn2023.0.3         amazonlinux           33 k
Installing weak dependencies:
 apr-util-openssl             x86_64          1.6.3-1.amzn2023.0.2          amazonlinux           15 k
 mod_http2                    x86_64          2.0.27-1.amzn2023.0.3         amazonlinux          166 k
 mod_lua                      x86_64          2.4.66-1.amzn2023.0.1         amazonlinux           60 k

Transaction Summary
================================================================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                          3.5 MB/s | 129 kB     00:00
(2/13): apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64.rpm                313 kB/s |  13 kB     00:00
(3/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                     1.9 MB/s |  97 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm             654 kB/s |  15 kB     00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm        694 kB/s |  19 kB     00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm                       1.6 MB/s |  47 kB     00:00
(7/13): httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch.rpm            571 kB/s |  13 kB     00:00
(8/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64.rpm                   30 MB/s | 1.4 MB     00:00
(9/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64.rpm                 2.4 MB/s |  81 kB     00:00
```

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

ec2-user@ip-172-31-45-237:~

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\thanv>cd downloads

C:\Users\thanv\Downloads>ssh -i nmindsproject001.pem ec2-user@16.171.154.239
The authenticity of host '16.171.154.239 (16.171.154.239)' can't be established.
ED25519 key fingerprint is SHA256:mwYvXnoCaSNuYxD7EMRZI98Ry4IDYbEfGO1DhKXARXE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '16.171.154.239' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-45-237 ~]$
```

It works!